

Logic

Exercises in modeling logic with binary variables. Notice that this is the **only** assignment in the course where you do not have to program in Julia/JuMP.

Problem 1 (OR)

Define constraints which relates the variables x and y to v such that v is 1 if *either* x or y are 1:

With three constraints:

- Enforce $v = 1$ if $x = 1$:

$$x \leq v$$

- Enforce $v = 1$ if $y = 1$:

$$y \leq v$$

- Enforce that $v = 0$ if neither $x = 1$ nor $y = 1$:

$$x + y \geq v$$

Alternatively with two constraints:

- Enforce $v = 1$ if $x = 1$ or if $y = 1$ (or both) using one collective constraint:

$$x + y \leq 2 \cdot v$$

- Enforce that $v = 0$ if neither $x = 1$ nor $y = 1$:

$$x + y \geq v$$

Problem 2 (AND)

Define constraints such that if both $x = 1$ *and* $y = 1$ then $v = 1$, otherwise $v = 0$:

- Enforce $v = 1$ if both $x = 1$ and $y = 1$:

$$x + y - 1 \leq v$$

- Enforce $v = 0$ if $x = 0$:

$$x \geq v$$

- Enforce $v = 0$ if $y = 0$:

$$y \geq v$$

Problem 3 (NOT)

Define one constraint such that if $x = 1$ then it forces $v = 0$ and vice versa so if $x = 0$ it forces $v = 1$.

- Enforce "not" relationship between x and v in one constraint:

$$v = 1 - x$$